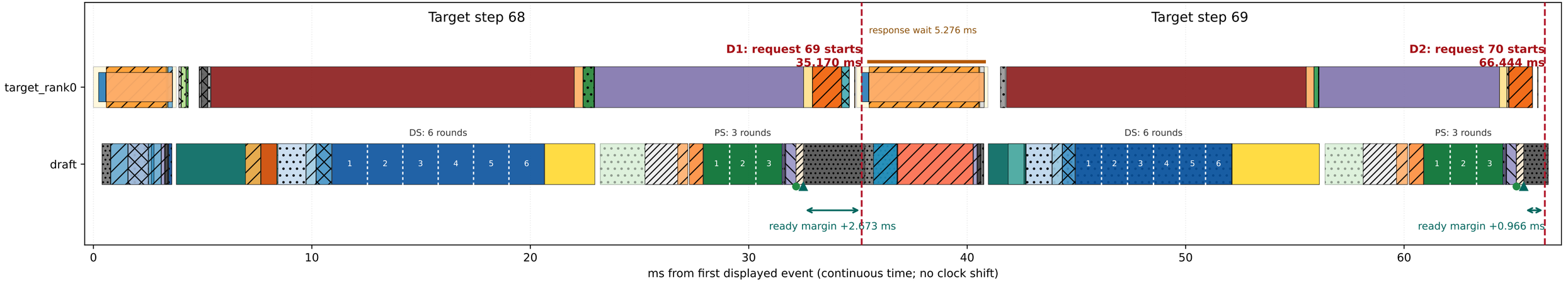
LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | P1 hit | UID 84_t0 | steps 68-69

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.

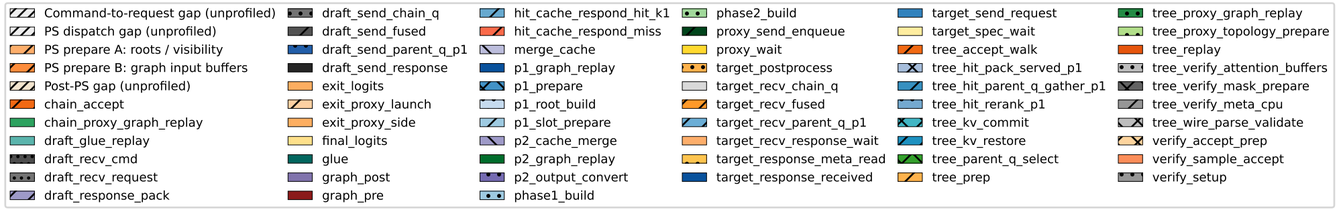


MEANS OVER ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	139	35.969	12.058	+2.389	1.584	9.352	+1.839	1.208
P2 hit	225	33.365	11.974	+1.955	1.593	9.390	+1.083	1.208
Miss	80	31.946	9.634	+4.198	1.211	9.095	+0.991	1.209
Overall	**444**	**33.924**	**11.579**	**+2.495**	**1.521**	**9.325**	**+1.303**	**1.208**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = finishes early; negative = late. Graph/K is an average per forward round INCLUDING sampling/selection, not isolated transformer latency.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

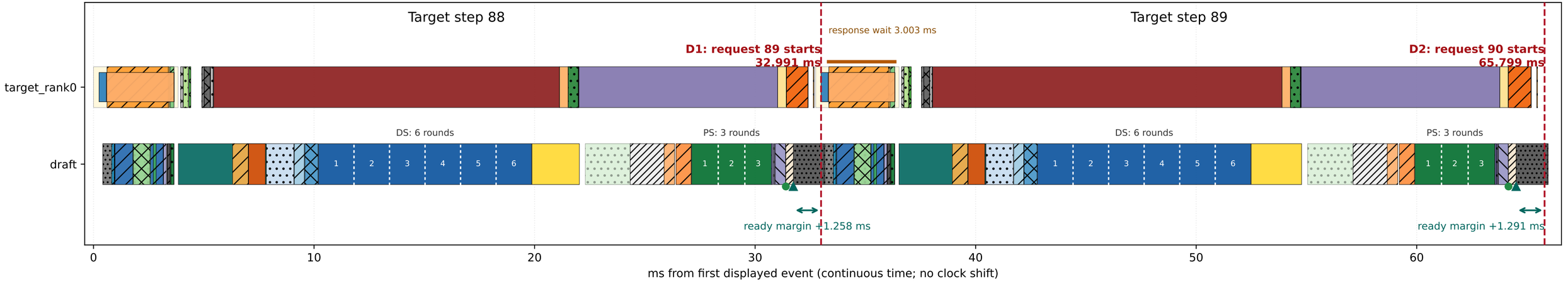
LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | P2 hit | UID 244_t0 | steps 88-89

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.

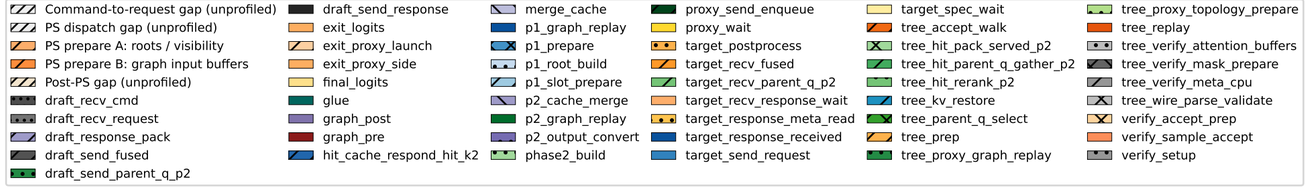


MEANS OVER ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	139	35.969	12.058	+2.389	1.584	9.352	+1.839	1.208
P2 hit	225	33.365	11.974	+1.955	1.593	9.390	+1.083	1.208
Miss	80	31.946	9.634	+4.198	1.211	9.095	+0.991	1.209
Overall	**444**	**33.924**	**11.579**	**+2.495**	**1.521**	**9.325**	**+1.303**	**1.208**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = finishes early; negative = late. Graph/K is an average per forward round INCLUDING sampling/selection, not isolated transformer latency.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

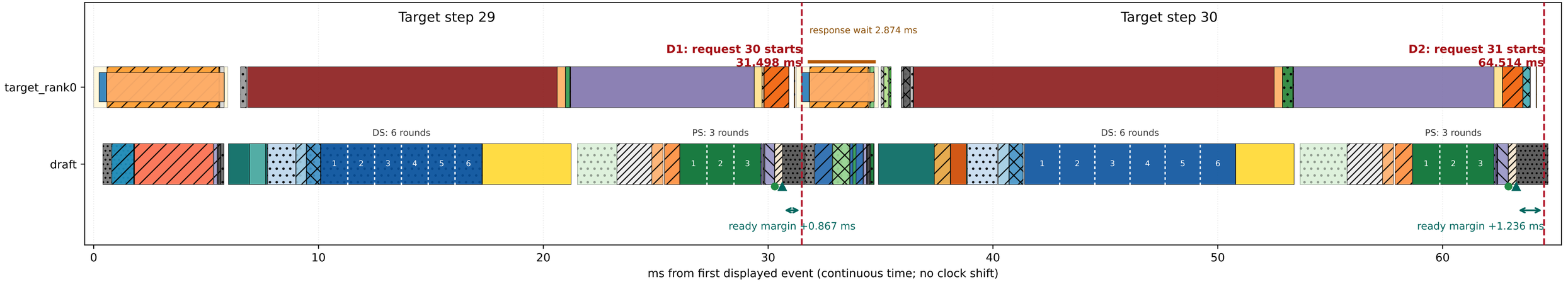
LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | Miss | UID 128_t0 | steps 29-30

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



MEANS OVER ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	139	35.969	12.058	+2.389	1.584	9.352	+1.839	1.208
P2 hit	225	33.365	11.974	+1.955	1.593	9.390	+1.083	1.208
Miss	80	31.946	9.634	+4.198	1.211	9.095	+0.991	1.209
Overall	**444**	**33.924**	**11.579**	**+2.495**	**1.521**	**9.325**	**+1.303**	**1.208**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = finishes early; negative = late. Graph/K is an average per forward round INCLUDING sampling/selection, not isolated transformer latency.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.